Supplementary Table 2 Secondary transmission or attack rates in School setting

Source	First author	Main issue	Study type	Population(N)	N primary/ index cases	N primary cases staff	N primary cases student	N Contact staff	N contacts student	N contacts	N contact tested	Setting	Comparator	Date of data collection	Outcome	N tested positive	Overall attack rate	Attack rate teacher	Attack Rate Student	Comments	Infection dynamics[1]	Incident cases and deaths per day[2]	Control measures
Attack Rate in School																							
Lancet Child and Adolescent Health	Macartney , K [30]	transmission of COVID-19 among children and staff in schools and early childhood education and care (ECEC) setting	Perspective cohort study	1475(27+1448)	27 (12 children and 15 adults)	15	12	NA	NA	1448	633	15 Schools and 10 ECEC in New South Wales Australia	students	25 January - 1 May, 2020	Secondary attack rate	**18**	**1.2 %**	**child to staff = 1%** **staff to staff = 4.4%**	**child to child = 0.3%**	**18 secondary** infections: 10 infections were from children and 8 infection were from adults.	**265.65** cases per million **3.61** deaths per million	**Incidence:** Start: 0.16 case/ M Peak= 19.49 cases/ M End: 0.47 cases/ M **Deaths** Start: 0.04 Peak= 0.31 End= 0	Physical distancing, hygiene measures and educational facility cleaning Online classes from 23rd March.
Euro Surveillance	Stein-Zamir , C [32]	COVID-19 outbreak in School in Israel. The school has grade 7-grade 12.	Observational	1314 (2+1312)	2	NA	2	151	1161	1312	1312	SARS-CoV-2 **outbreak** in school of Israel after school reopening.	Students	May-June, 2020	attack rate	**178 (Student= 153 and Staff= 25)**	**13.5 %**	**16.6 %**	**13.2%**	The number of cases was higher in lower grades (grade 7-9)	1089.61 cases per million 11.32 death per million	**Incidence** Start =13.40 Peak= 90.81 End= 90.81 **Deaths** Start= 0.46 Peak= 0.69 End= 0.12	Hygiene, facemasks, social distancing and minimal interaction on between classes.

[1] Total number of confirmed COVID-19 cases and deaths from COVID-19 during the data collection/ study duration in the country of study
[2] The number of incidence case per million and number of death per million per day. The data presented are at the start of the study period, peak during the study period and at the end of the study period.

Euro Surveillance	Larosa, E [40]	Secondary transmission of COVID-19 cases in school and preschool setting of northern Italy	Observational	1248 (48+1200)	48	5	43	204	996	1200	1198 (994 student & 204 staff)	Total 36 different educational setting (school, preschool and infant toddler center) in northern Italy	Students of different school level	1 September- 15 October, 2020	attack rate	**38**	**3.2 %**	**0**	**Overall = 3.8% Secondary School AR= 6.6% Elementary school AR= 0.38% Preschool AR= 0**	All the secondary cases were among the students who were tested	1842.70 cases per million 14.75 deaths per million	**Incidence** Start= 16.13 Peak= 145.6 End= 145.60 **Deaths** Start= 0.13 Peak= 1.37 End= 1.37	Physical distancing measures, mandatory wearing of surgical masks for children, single desk and desk must me 1 m apart, extracurricular activities were suspended.
Frontiers of Public Health	Hernandez, A [29]	Secondary transmission from the school staff	Case report	202(2+200)	2	2	NA	NA	NA	200 approximately	20 (10% of the contact)	School of Panama	Students	February- March, 2020	attack rate	**7**	**3.5 %**	**3.5%**	**0**	All 15 close contacts of teachers tested negative for SARS-CoV-2 **Positive cases were the school contacts**	273.48 cases per million 6.72 deaths per million	**Incidence** Start= 0.23 Peak= 44.50 End= 44.50 **Death** Start= 0.23 Peak= 1.62 End = 1.39	School closure from March
Euro Surveillance	Heavey, L [42]	Secondary transmission in school setting	Observational	1031 (6+1025)	6	3	3	NA	NA	1025	NA	School in Ireland	NA	12 March 2020	Secondary transmission, SAR	**0 in School** **2 outside school**	**0**	**0**	**0**	**2 contacts of adult cases outside of school developed SARS-**	8.71 cases million 0.2 deaths per million	No cases and no deaths on the particular day in	School closure from 12th March

																			CoV-2 infection with overall AR 0.17%		Ireland	
Clinical Infectious Diseases	Yung, CF [34]	Secondary transmission in school setting	Observational	122 (3+119)	3	1	2	NA	NA	119	119 (8+34 +77)	3 different school in Singapore	NA	February-March 2020	Secondary transmission, AR	**0**	**0**	**0**	**0**	No secondary cases were found among close contacts.	**155.55 cases per million** **0.17 deaths per million**	**Incidence** Start= 0.51 Peak= 12.48 End= 8.03 **Deaths** Start= 0.34 Peak= 0.34 End= 0	Terminal cleaning of school, suspension of extracurricular activities, preschool was closed for 14 days
Clinical Infectious Diseases	Danis, K [27]	Tertiary transmission in school setting	Observational	113 (1+112)	1	NA	1	NA	NA	112	101	3 schools and 1 ski club in French Alps	Student and teachers	February 2020	Tertiary transmission	**0**	**0**	**0**	**0**	No transmission in schools	1.44 cases per million 0.03 deaths per million	**Incidence** Start= 0.02 Peak= 0.31 End= 0.29 **Deaths** Start= 0.02 Peak= 0.02 End= 0	**Use of surgical masks**
National Centre for Immunisation Research and Surveillance, NSW, Australia	NCIRS [35]	secondary transmission in differed educational setting	Observational	527 (6+521)	6	2	4	62	459	521	319	6 different educational (5 schools and 1 ECEC)	Student; teacher	10 April – 3 July, 2020	Secondary transmission	**0**	**0**	**0**	**0**	No secondary transmissions	80.20 cases per million 1.96 deaths per million	**Incidence** Start= 4.20 Peak= 7.61 End=7.61 **Deaths** Start= 0.12 Peak= 0.31 End= 0	**Schools** were re-opened, full face-to-face teaching commenced while implementing COVID-19 safe practice

National Centre for Immunisation Research and Surveillance, NSW, Australia	NCIRS [38]	secondary transmission in differed educational setting	Observational	3863(33+3824)	39	7	32	385	3439	3824	3641	34 different educational setting in NSW which includes 28 schools and 6 ECEC	student	4 July – 25 September, 2020	Secondary transmission; attack rate	**33**	**0.86 %**	**1.29 %**	**0.81%**	Among tested: AR= 0.9% High School AR= 1.1% Primary school AR= 0.4% ECEC AR= 0.7%	728.36 cases per million 30.04 deaths per million 744 cases across NSW. Among them 555 were locally acquired.	**Incidence** Start= 7.18 Peak= 28.08 End= 0.63 **Death** Start= 0 Peak= 2.31 End= 0.04	Schools were open for face-to-face learning while implementing COVID-19 safe practice.
Lancet Infectious Diseases	Ismail, SA [36]	estimating the rate of SARS-CoV-2 infection and outbreaks among staff and students in educational settings during the summer half-term	Prospective cross sectional analysis	928000 (Students only)	55	34	21	NA	NA	NA	NA	Schools in England	Student	June/July 2020	Secondary transmission	**NA**	**122**	**100**	**22**	Considering only outbreak situation there were 122 secondary cases in school during this period. Among them 100 cases were staff and 22 were students	661.80 cases per million 53.63 deaths per million	**Incidence** Start= 21.27 Peak= 21.65 End= 13.01 **Death** Start= 1.27 Peak= 3.80 End= 0.29	School closed from March 20 and reopened from June 1st after implementing smaller classes separated into distant social bubbles
Morbidity and Mortality	Falk, A [39]	Secondary transmission in	Observational	5530 (184+5346)	184	58	126	596	4750	5346	5530	Schools in Wiscon	Student	August-29 November,	Secondary transmission;	**191**	**0.13 %**	**0**	**0.14%**	Among 191 tested positive,	26,802.94 cases	**Incidence**	Compulsory masks for

Weekly Report		school setting										sin, USA		2020	attack rate					7 were secondary cases There was no secondary transmission among staff	per million 339.08 deaths per million	Start= 169.71 Peak= 628.19 End= 423.22 **Deaths** Start= 3.31 Peak= 6.86 End= 3.13	student and staff. Classroom size 11-20 students and indoor lunch.